

The microbiome and obesity: is obesity linked to our gut flora?

The human gut is a lush microbial ecosystem containing about 100 trillion microorganisms, whose collective genome, the microbiome, contains 100-fold more genes than the entire human genome. The symbiosis of our extended genome plays a role in host homeostasis and energy extraction from diet. In this article, we summarize some of the studies that have advanced the understanding of the microbiome and its effects on metabolism, obesity, and health. Metagenomic studies demonstrated that certain mixes of gut microbiota may protect or predispose the host to obesity. Furthermore, microbiota transplantation studies in germ-free murine models showed that the efficient energy extraction traits of obese-type gut flora are transmissible. The proposed methods by which the microbiome may contribute to obesity include increasing dietary energy harvest, promoting fat deposition, and triggering systemic inflammation. Future treatments for obesity may involve modulation of gut microbiota using probiotics or prebiotics.